

9.3 Assessment criteria

A: Summary of case study or physics-based visit

Ref	Criterion	Mark
S1	Carries out a visit OR uses library, consulting a minimum of three different sources of information (eg books/websites/journals/magazines/case study provided by Edexcel/manufacturers' data sheets)	1
S2	States details of visit venue OR provides full details of sources of information	1
S3	Provides a brief description of the visit OR case study	1
S4	Makes correct statement on relevant physics principles	1
S5	Uses relevant specialist terminology correctly	1
S6	Provides one piece of relevant information (eg data, graph, diagram) that is not mentioned in the briefing papers for the visit or case study	1
S7	Briefly discusses context (eg social/environmental/historical)	1
S8	Comments on implication of physics (eg benefits/risks)	1
S9	Explains how the practical relates to the visit or case study	1
	Maximum marks for this section	9

B: Planning

Ref	Criterion	Mark
P1	Lists all material required	1
P2	States how to measure one relevant quantity using the most appropriate instrument	1
P3	Explains the choice of the measuring instrument with reference to the scale of the instrument as appropriate and/or the number of measurements to be taken	1
P4	States how to measure a second relevant quantity using the most appropriate instrument	1
P5	Explains the choice of the second measuring instrument with reference to the scale of the instrument as appropriate and/or the number of measurements to be taken	1
P6	Demonstrates knowledge of correct measuring techniques	1
P7	States which is the independent and which is the dependent variable	1
P8	Identifies and states how to control all other relevant variables to make it a fair test	1
P9	Comments on whether repeat readings are appropriate in this case	1
P10	Comments on safety	1
P11	Discusses how the data collected will be used	1
P12	Identifies the main sources of uncertainty and/or systematic error	1
P13	Draws an appropriately labelled diagram of the apparatus to be used	1
P14	Plan is well organised and methodical, using an appropriately sequenced step-by-step procedure	1
	Maximum marks for this section	14

C: Implementation and Measurements

Ref	Criterion	Mark
M1	Records all measurements using the correct number of significant figures, tabulating measurements where appropriate	1
M2	Uses correct units throughout	1
M3	Obtains an appropriate number of measurements	1
M4	Obtains measurements over an appropriate range	1
	Maximum marks for this section	4

D: Analysis

Ref	Criterion	Mark
A1	Produces a graph with appropriately labelled axes and with correct units	1
A2	Produces a graph with sensible scales	1
A3	Plots points accurately	1
A4	Draws line of best fit (either a straight line or a smooth curve)	1
A5	Comments on the trend/pattern obtained	1
A6	Derives relation between two variables or determines constant	1
A7	Discusses/uses related physics principles	1
A8	Attempts to qualitatively consider sources of error	1
A9	Suggests realistic modifications to reduce error/improve experiment	1
A10	Calculates uncertainties	1
A11	Provides a final conclusion	1
Maximum marks for this section		11

E: Report

Ref	Criterion	Mark
R1	Summary contains few grammatical or spelling errors	1
R2	Summary is structured using appropriate subheadings	1
Maximum marks for this section		2

Total marks for this unit		40
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